

PROJECT EXPO 2026

Smart Hospital Care System: IOT-Based Patient Monitoring and Alert System

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PROBLEM STATEMENT:

Hospitals require continuous monitoring of IV fluid levels and patient bed occupancy to ensure timely care. Manual monitoring can lead to delayed detection of low IV fluid levels or unexpected patient bed exit, especially when nurses are handling multiple patients. Therefore, there is a need for an embedded IoT system that automatically monitors these parameters, provides early alerts, and prioritizes critical situations to support faster nurse response.

ABSTRACT:

The Smart Hospital Care System is an embedded IoT-based solution designed to improve patient safety through continuous monitoring of IV fluid levels and patient bed occupancy. An STM32 microcontroller collects and processes sensor data to determine the remaining IV fluid, flow rate, estimated time-to-empty, and patient bed status. An ESP32 provides wireless communication between the system and a nurse monitoring dashboard.

The system generates alerts for low, near-empty, and empty IV fluid conditions, as well as unexpected patient bed exit. These alerts are organized using an intelligent nurse priority queue based on their severity, helping nurses respond to critical situations first. A patient admission control mechanism ensures that monitoring and alerts are activated only for admitted patients.

The proposed system reduces manual monitoring, enables faster response to important events, and provides a practical and affordable approach for continuous patient-care monitoring in hospitals.

OBJECTIVES:

- To monitor the IV fluid level continuously using a load cell and STM32.
- To calculate the remaining IV fluid, flow rate, and estimated time-to-empty.
- To monitor patient bed occupancy and detect unexpected bed exit using FSR sensors.
- To generate real-time alerts for low IV fluid, near-empty IV, empty IV, and bed exit conditions.
- To implement a priority queue that helps nurses respond to critical alerts first.
- To transmit the processed data wirelessly using ESP32 to a nurse monitoring dashboard.
- To reduce manual monitoring and improve the response time of healthcare staff.

SCOPE OF THE PROJECT :

The Smart Hospital Care System is designed to assist nurses in continuously monitoring IV fluid status and patient bed occupancy. The system can be used in hospitals, wards, ICUs, emergency areas, and other healthcare environments.

The system includes:

- IV fluid level and remaining-time monitoring.
- IV flow rate calculation.
- Patient bed occupancy monitoring.
- Unexpected bed-exit detection.
- Real-time alerts and notifications.
- Intelligent nurse priority queue.
- Wireless monitoring through an ESP32-based dashboard.
- Patient admission control to activate monitoring only for admitted patients.

FUTURE ENHANCEMENTS:

- Heart rate and other vital-sign monitoring.
- ECG integration.
- Cloud-based patient data storage.

- Hospital Information System (HIS) integration.
- Voice-based alerts.
- AI-based anomaly detection.

EXPECTED OUTCOMES:

- Continuous monitoring of IV fluid level and patient bed occupancy.
- Accurate calculation of remaining IV fluid, flow rate, and estimated time-to-empty.
- Automatic detection of low, near-empty, and empty IV conditions.
- Detection of unexpected patient bed exit.
- Automatic prioritization of critical alerts using a nurse priority queue.
- Wireless transmission of patient and alert information to the nurse dashboard.
- Reduced manual monitoring workload for nursing staff.
- Faster response to critical patient-care situations.
- Improved patient safety and efficiency of hospital monitoring.

LITERATURE SURVEY:

AUTHOR	CONTRIBUTION	LIMITATION
Ignacio Gheresi et al. (2018)	Studied smart-bed technologies for monitoring patients and improving safety in healthcare environments.	Mainly focuses on smart-bed functions and does not include IV fluid monitoring.
Alexandre Andrade et al. (2024)	Reviewed IoT-based healthcare systems for continuous patient monitoring and remote healthcare applications.	Mainly focuses on patient monitoring and does not specifically address IV fluid level and bed-exit detection together.
Guruprakash K. S. et al. (2025)	Proposed an IoT-enabled smart hospital framework for improving healthcare monitoring and management.	Does not specifically combine IV fluid estimation, bed occupancy monitoring, and alert prioritization.
Jan Velicka et al. (2025)	Developed an IoT-based monitoring approach with remote sensing and alert generation for healthcare applications.	The system is focused on specific monitoring parameters and does not provide integrated IV and patient-bed monitoring.
Yiting Bai et al. (2025)	Proposed an intelligent ICU monitoring approach for continuous patient observation and early detection of critical conditions.	Requires more complex AI-based infrastructure and focuses mainly on patient condition rather than hospital resource monitoring.

EXISTING SOLUTION ANALYSIS:

Feature	Existing Solutions	Proposed System
IV fluid monitoring	Mostly manual or separate monitoring systems	Automatic monitoring using load cell and STM32
Remaining IV fluid	Manually estimated	Automatically calculated
IV flow rate	Usually not monitored continuously	Continuously calculated
IV time-to-empty	Usually not available	Estimated based on fluid usage
Bed occupancy	Manual observation or separate systems	Automatically detected using FSR sensors
Unexpected bed exit	Limited or separate monitoring	Automatically detected and alerted
Alert generation	Manual or basic threshold alerts	Automatic real-time alerts
Alert prioritization	Usually handled manually	Nurse Priority Queue
Wireless monitoring	Limited in conventional systems	ESP32-based wireless dashboard
Patient admission control	Generally manual	Monitoring activated for admitted patients
System architecture	Separate monitoring systems	STM32 + ESP32 integrated system
Main purpose	Individual parameter monitoring	Combined IV and bed monitoring with intelligent alert management

ADVANTAGES OF THE PROPOSED SYSTEM:

- Reduces dependence on continuous manual monitoring.
- Provides real-time IV and bed status.
- Gives early warnings for low and critical IV conditions.
- Detects unexpected patient bed exit.
- Prioritizes important alerts for faster nurse response.
- Provides wireless monitoring through a centralized dashboard.
- Uses low-cost embedded hardware suitable for practical healthcare applications.

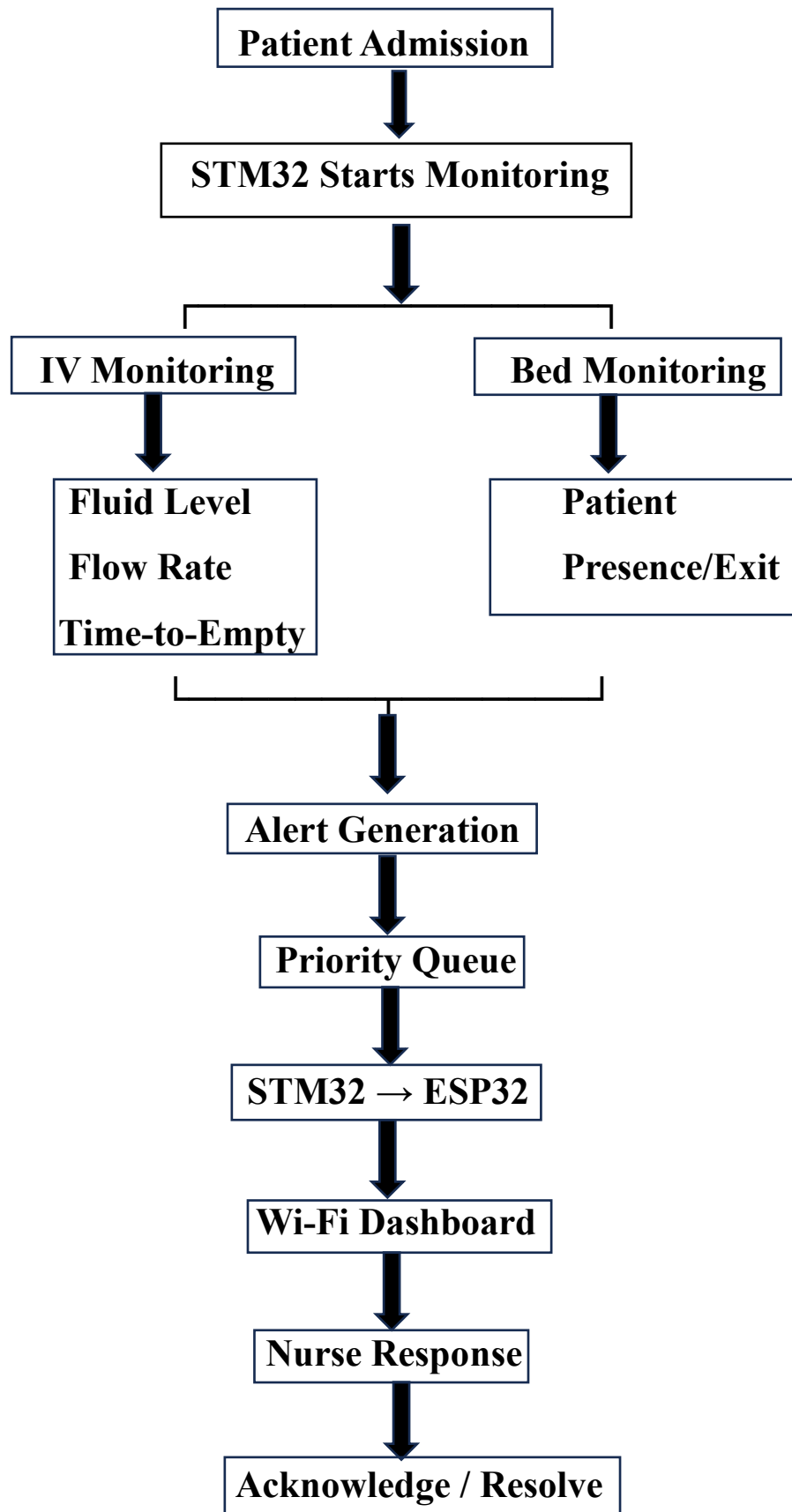
GAP ANALYSIS:

Parameter	Existing Systems	Gap Identified	Proposed System
IV fluid monitoring	Several systems monitor IV fluid using load cells or other sensors.	Usually focused only on IV monitoring.	Combines IV monitoring with patient bed monitoring.
IV remaining quantity	Fluid level/weight can be measured automatically.	Some systems provide only level/threshold alerts.	Calculates remaining fluid and percentage.
IV time-to-empty	Available in some advanced IV monitoring systems.	Often not integrated with other patient-care events.	Calculates flow rate and estimated time-to-empty.
Bed occupancy	Smart-bed systems can monitor patient presence.	Usually developed as a separate system.	Integrates bed occupancy with IV monitoring.
Bed-exit detection	Available in some smart-bed solutions.	May require dedicated smart-bed infrastructure.	Uses FSR sensors with STM32 for bed status and exit detection.
Alert generation	Usually parameter-specific alerts.	Multiple alerts may require separate monitoring systems.	Provides unified IV and bed alerts.

Alert prioritization	Often dependent on manual nurse response.	No simple integrated priority mechanism for the monitored parameters.	Uses a nurse priority queue based on alert severity.
Wireless monitoring	IoT-based IV and smart-bed systems provide remote monitoring.	Monitoring functions are generally distributed across separate systems.	ESP32 provides a common wireless dashboard.
Patient admission	Not always linked to monitoring logic.	Alerts may be generated even when a bed is not assigned to a patient.	Patient admission control activates monitoring for admitted patients.
Overall integration	Individual solutions are available.	Limited integration of IV status, bed status and alert priority in one low-cost prototype.	Integrates all required functions into one embedded IoT system.

NOVEL FEATURES:

- Integrated IV and bed monitoring in one system.
- Automatic IV fluid level and time-to-empty calculation.
- Real-time patient bed occupancy and exit detection.
- Priority-based nurse alert queue for critical situations.
- Patient admission-based alert control.
- Wireless nurse dashboard using ESP32.
- STM32-based real-time sensor processing.
- Low-cost and practical healthcare monitoring solution.

Proposed System Workflow:

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